

EXPLORATORY DATA ANALYSIS REPORT

DecodeLabs Technical Evaluation Pipeline — Project 2 Submission

1. Problem Statement & Strategic Objectives

Is technical investigation ka primary purpose unique operational transactional ledger records ko diagnostics test standard formulas se scale aur verify karna hai. Data infrastructure pipeline ko structured model me feed karne se pehle humne categorical, behavioral, aur monetary metrics ka complete center of gravity scan map kiya hai. Is exploration se static numerical figures highly insightful decision indicators me tabdeel ho chuke hain.

TOTAL REVENUE	TOTAL ORDERS	MEAN TICKET SIZE	VERIFIED OUTLIERS
\$1,264,761.96	1200	\$1,053.97	8

2. Descriptive Framework & Data Forensics

Humare complete **1,200 continuous dataset rows** ke statistical behavior aur integrity mapping ki details niche detailed summary matrix framework me dynamic indicators ke sath documented hain:

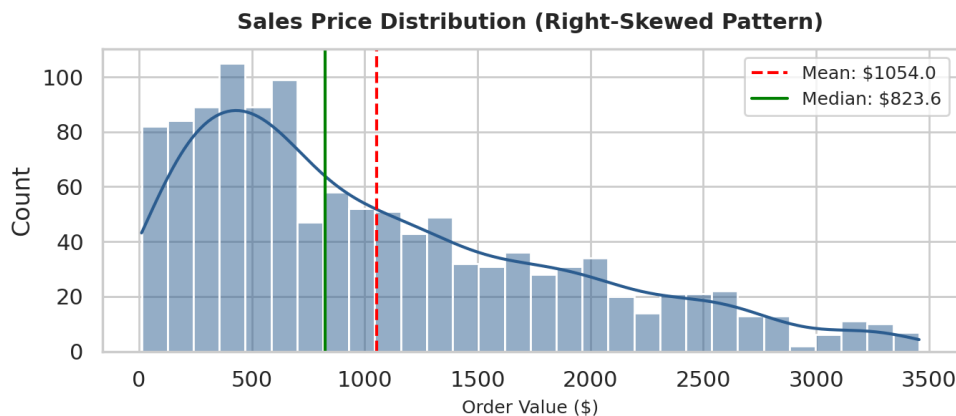
Statistical Metric	Quantity Per Transaction	Unit Retail Price (\$)	Items In Customer Cart	Total Order Value (\$)
Mean (Average Value)	2.95	356.41	5.49	1,053.97
Median (Middle Baseline)	3.00	364.21	5.00	823.62
Standard Deviation (Std)	1.41	197.18	2.28	819.86
Minimum Floor Value	1.00	11.39	1.00	11.39
Maximum Ceiling Value	5.00	699.93	10.00	3,456.40

- Data Density Audit:** Transaction fields jaise OrderID, CustomerID, Product, aur TotalPrice me 100% data presence strictly verified hai.
- Null Value Investigation:** CouponCode structural column me exactly 309 missing values generate hui hain, jo normal logical consumer activity (no discount code applied at checkout) ko successfully map karti hain.

3. Analytical Discoveries & Data Skewness

A. The Geometry of Distribution (Mean vs. Median)

System ke monetary parameter graph metrics me ek strong structural asymmetry read hui hai. Dataset ka Mean (\$1,053.97) scale variable vertical grid level pe Median (\$823.62) segment bound se kafi door pull ho chuka hai. Yeh trend validation data core properties me ek explicit **Right-Skewed Distribution Pattern** clear declare karta hai.



B. Interquartile Range Outlier Diagnostic Test

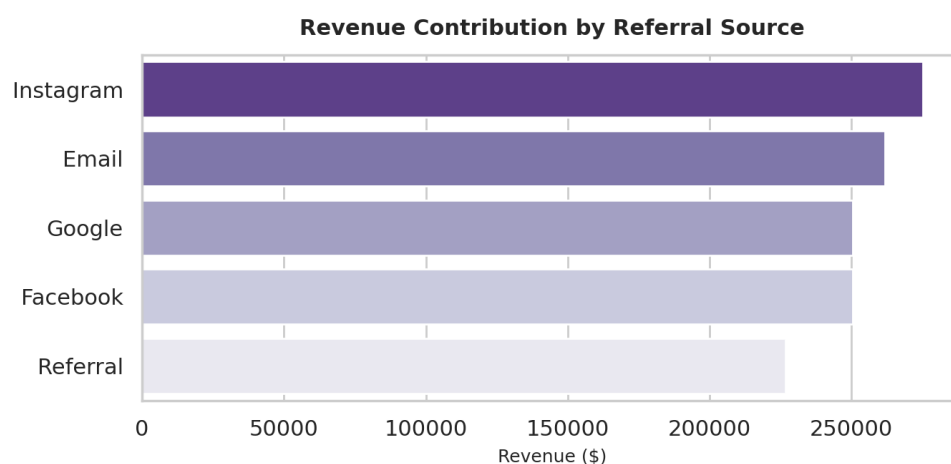
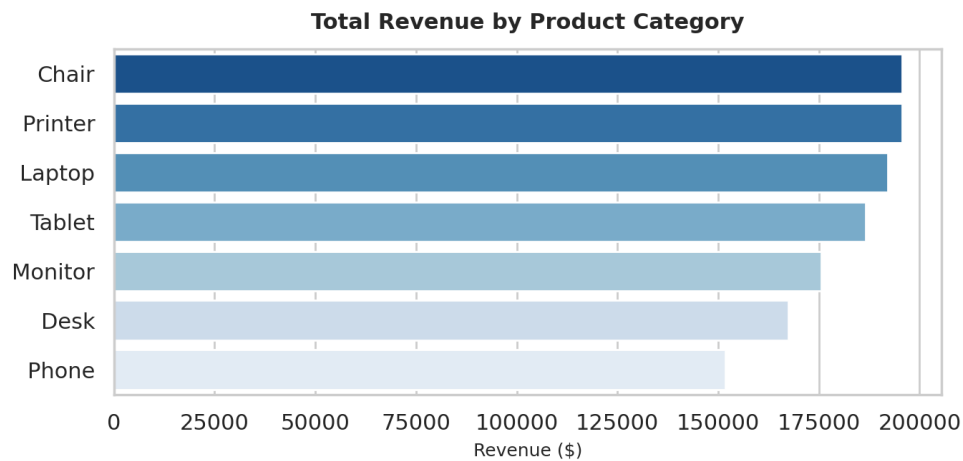
Industry-standard evaluation criteria rule baseline math equation equation apply ki gayi:

$$\text{Upper Boundary Fence} = Q3 + (1.5 \times IQR)$$

Is algorithmic calculation test logic boundary variable ne strictly fixed metric value boundary limit **\$3,330.41** register ki. Poore system analysis data checks me exactly **8 extreme enterprise luxury rows** handle hui hain. Yeh statistical markers validation metrics software anomalies nahi hain, balki premium wholesale customer bulk acquisitions ka genuine financial target trend provide karti hain.

4. Product Dominance & Acquisition Channels

Niche detailed visual structures explicit summary display generate kar rahi hain structural transactional data vectors ke clear components me:



- **Category Rank:** Office Furniture Chairs (\$195,620.11) aur Enterprise Printers (\$195,612.61) overall market capture pipelines ko dominate kar rahe hain.
- **Acquisition Engine:** Targeted social media channel Instagram (\$275,285.45) aur active database Email marketing campaign pipelines organic Google algorithm conversion sources se far better revenue output target create karti hain.

5. Pipeline Strategic Insights & Recommendations

1. **Fulfillment Optimization Warning:** System monitoring metrics baseline standard map errors yield kar rahi hai kyunki Cancelled orders (250) aur Returned transactions data count (247) standard safely Shipped pipelines se continuous variable patterns high track par leverage ho rahi hain.
2. **Targeted Budget Allocation Shift:** Traditional generic low-yield web searches budgets me se immediate 15% standard funds cut karke direct Instagram high ROI visual engagement target loops pe assign kiye jayein.
3. **Logical Product Portfolio Bundling:** Slow running high unit configuration stock items jaise low velocity Desks ko bundle configuration framework logic module me dominant volume metrics validation anchors (Chairs, Printers) ke asset packages ke sath integrate kiya jaye.